

The Development of Selective Attention: Holistic Perception versus Resource Allocation

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In two experiments, a speeded sorting task was administered to children. In each sorting, the child was instructed to sort a deck of cards according to values on one dimension while the values of a second irrelevant dimension were held constant, were correlated, or were varied orthogonally. Experiment 1 evaluated developmental changes in perceived structure as compared with such changes in the control of attention. Different groups of kindergarten, 2d-, and 5th-grade children performed the task with spatially integrated vs spatially separated dimensions. The results indicate that with spatially integrated dimensions there is a trend from holistic to featural perception and an increase in attentional control with increasing age. Spatially separate dimensions are perceived dimensionally at all ages, but show a trend in attentional development. In Experiment 2, the spatially integrated dimensions of Experiment 1 were made separable by portraying different targets within a configuration in different colors. The theoretical implications of the results are discussed. © 1987 Academic Press, Inc.

One of the most thoroughly documented findings in the developmental literature is that older children perform more effectively than younger children in tasks where successful performance requires that irrelevant sources of information be ignored. This trend has been revealed in a variety of tasks including selective listening (e.g., Doyle, 1973; Maccoby,

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1969), speeded classification (e.g., Pick & Frankel, 1973; Smith, Kemler, & Aronfreed, 1975; Strutt, Anderson, & Well, 1975; Well, Lorch, & Anderson, 1980), incidental learning and memory (e.g., Kemler, Shepp, & Foote, 1976; Hagen & Hale, 1973), and some forms of classification (e.g., Bruner, Olver, & Greenfield, 1966; Wohlwill, 1962).

This developmental trend has suggested two types of interpretation. The first, and more traditional, argues for a developmental difference in attention. According to this view, the younger child focuses attention less well than does the older. Thus, the younger child may have difficulty in focusing on the relevant aspects of a task and in filtering out irrelevant sources of information as well. With increasing age and experience, however, the child is assumed to gain increasing command over attentional resources (e.g., Pick & Frankel, 1973; Hagen & Hale, 1973).

The second and more recent view borrows from the work of Garner (1974, 1976) and Lockhead (1972) and argues that age-related differences in performance may be due to developmental differences in perceived structure. The evidence that supports this interpretation comes from speeded sorting (Shepp & Swartz, 1976) and restricted classification tasks (Smith & Kemler, 1977), and strongly suggests that younger children perceive some multidimensional objects as integral wholes, whereas older children perceive the same objects as conjunctions of separable features. Since selective attention to the features of an integral stimulus is not possible (e.g., Garner, 1974), developmental performance trends that appear in selective attention tasks may be due to developmental differences in perceived structure rather than to developmental differences in attention.

A fundamental assumption of this recent work on the development of perceived structure (e.g., Shepp, 1977, 1978; Smith & Kemler, 1977, 1978) is that age-related differences in performance may be jointly affected by developmental differences in perceived structure as well as by attentional control. If sources of information are perceived as independent by younger and older children alike, then performance differences in a selective attention task can be attributed to developmental differences in attentional control. If, however, the sources of information that are perceived as separable features by older children are perceived as integral wholes by younger children, then age-related differences in performance will reflect both developmental differences in perceived structure and attention.

According to this view, younger children may fail to perform selective attention tasks successfully either because they perceive multidimensional objects as integral wholes or because they do not allocate attentional resources effectively to independent sources of information. Older children, on the other hand, should successfully perform selective attention tasks if the sources of information are independent, and should fail only if the sources of information interact.

An experimental test for multiple trends in perceptual developmental

has two requirements. First, there must be one task to assess developmental differences in attention and another to assess such differences in perceived structure. Second, it must be possible to present within each of these tasks one dimensional combination that is perceptually independent at all ages and another where perceptual independence is associated with increases in age.

A task that fulfills the requirement is speeded sorting. In this task the subject is given a deck of cards and instructed to sort them into two piles as quickly as possible. In different conditions of the experiment, the subject sorts on the basis of one dimension while a second either is held constant, presents values that are correlated with values on the target dimension, or presents values that are varied orthogonally relative to the values on the target dimension.

The dimensional combinations that satisfy our requirement are displayed on the stimuli in Fig. 1, which have three dimensions. The dimensions are the number of lines in the outer ring (10 vs 15), the orientation of pointers (330° vs 30°), and the color of the inner configuration (red vs red-orange). From these stimuli, two sets of dimensional combinations were arranged. The number of lines and color of the inner disc were combined in one set (the pointer was removed), whereas the orientation of the pointer and the color of the inner configuration were combined in the second (the lines were removed).

A priori determinations of perceptual independence or separability are difficult to make. Often dimensional combinations that are integral according to definitions offered by Garner (1976) or Lockhead (1966a) are separable when performance criteria are used. Spatial separation of dimensions, however, is an operation that frequently produces perceptual independence (Lockhead, 1966b). Garner and Felfoldy (1970) have shown, for example, that chroma and value (ordinarily integral dimensions) become separable

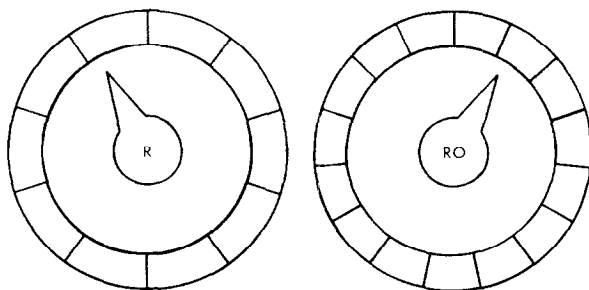


FIG. 1. An illustration of the dimensions in Experiments 1 and 2. The spatially separated dimensions were the number of lines in the outer ring vs the color of disc (pointer was removed). The spatially integral dimensions were color vs angle of pointer (outer ring and lines were removed). In Experiment 2, the color of the disc vs the angle of the pointer (the pointer was always black) were the dimensions.

if the variations on each of these dimensions are displayed at spatially separated locations. By hypothesis, the number of lines vs color are also separable and are so perceived regardless of age. In comparison, spatially integral dimensions such as orientation and color are separable for the adult and integral for young children.¹

The perceptual independence between dimensions is inferred by comparing the performances in the control task with those in which the values of the target and nontarget dimensions are correlated. Because the perceived differences between stimuli that are generated by combining integral dimensions are perceived by overall similarity relations (e.g., Handel & Imai, 1972), the redundancy introduced with correlated dimensions serves to make the stimuli more discriminable than stimuli in the control task. Consequently, correlated decks are sorted faster than control decks. In contrast, the perceived differences between stimuli that are generated by combining separable dimensions are perceived according to dimensional relations (e.g., Handel & Imai, 1972; Shepard, 1964), and the redundancy that is added to the stimuli by correlated values does not modify their discriminability relative to the stimuli in the control task. With separable stimuli, the correlated dimensions task is performed no faster than the control task (e.g., Garner & Felfoldy, 1970).

Differences in selective attention are inferred by comparing performances on the orthogonal dimensions and control tasks. For integral dimensions, with their underlying similarity structure, selective attention to features is not possible. As a consequence, it should take longer to sort in an orthogonal dimensions task than in a control task. With separable dimensions, however, selective attention to specific features is possible, and subjects should be able to sort in the orthogonal dimensions task as quickly as in a control task.

The present experiment investigates multiple trends in perceptual development by having children of different ages sort cards in a speeded task in which the stimuli are either spatially separated or integrated. The predictions are straightforward. By hypotheses, the number vs color dimension should be separable at all developmental levels, and no redundancy gains in tasks with correlated values would be expected. Interference effects would be expected to occur, however, with orthogonal dimensions. The magnitude of these effects should vary inversely with age, reflecting a developmental trend toward improving attentional skills. By contrast, the orientation vs color dimensions should yield different perceived structures at different ages. The stimuli generated by these

¹ A pilot study using a restricted classification task did reveal overall similarity classifications for kindergartners and dimensional classifications for adults for angle vs color. Adults also classified lines vs color by dimension, but kindergartners classified exclusively by one dimension, i.e., they centrated.

dimensions should behave as integral for young children and as separable for older children. Thus, redundancy gains with correlated values should occur in the young child's performance, but such gains should attenuate with increasing age. Interference effects with orthogonal dimensions should also appear in the young child's performance, and these effects should also decline with increasing age.

EXPERIMENT 1

Method

Subjects. The subjects were children who were attending school in East Providence, Rhode Island. There were 48 kindergartners (mean age 5 years 2 months; range 4 years 11 months to 5 years 11 months), 48 second-graders (mean age 7 years 5 months; range 7 years to 8 years 1 month), and 48 fifth-graders (mean age 11 years 6 months; range 11 years 1 month to 12 years 2 months). The subjects were assigned randomly to all treatment groups with the constraint that groups were balanced as closely as possible for sex.

Stimuli. The orientation of the pointer and the color of the figure comprise the spatially integrated dimensions; the number of lines in the outer ring and the color of the inner disc comprise the spatially separated dimensions (see Fig. 1). For each type of dimensional combination, both hard and easy stimulus sets were created by changing the discriminability of both dimensions. For the spatially integrated dimensions, the orientation values chosen were 45° vs 315° and 30° vs 330° when measured from the vertical axis. The figures were cut from Color Aid paper; the colors were red vs red-orange and red vs red-orange-red. To create the spatially separated stimuli, a 1.9-cm circle was cut from the same shades of Color Aid paper. The outer ring was drawn in black ink using two concentric circles, 7.6 and 10.2 cm in diameter. The number of lines in the outer ring were 5 vs 15 and 10 vs 15. For each value, the lines were arranged in three different configurations to ensure that subjects were using number and not general orientation. All stimuli were constructed on 12.7-cm square white cards, and the levels of discriminability used were chosen from a scaling study in which adult subjects made dissimilarity judgments.

Task. A speeded sorting task similar to that of Shepp and Swartz (1976) was used. The subject was instructed to sort a deck of 32 cards into two piles. Three types of decks were employed. In control decks, the values on one dimension were varied and the values of the second dimension were held constant. In the correlated deck, the values on both dimensions were varied, and a specific value on one dimension was consistently paired with a specific value on the other. In orthogonal decks, the values of both dimensions were again varied but both values of one dimension were paired with both values on the other, yielding four distinctive stimuli.

Procedure. The subjects were tested individually in a small, quiet room at school. The subjects were told that they were going to play a game in which they were to sort a deck of cards into two piles as quickly as possible without making errors. Depending upon the stimulus set to which they were assigned, subjects were told the specific stimuli that belonged in the alternative classifications for each of the types of decks. For example, with the color and orientation set, subjects were told that sometimes color would be the target dimension, and the stimuli that looked one way (e.g., red) should go in one pile, whereas cards that looked another way (e.g., orange) would go in the other pile. Similarly, when orientation was the relevant dimension subjects were told that stimuli that pointed one way (e.g., 330°) should go in one pile while those that pointed the other way (e.g., 30°) would go in the other. At the beginning of each sort, the child was shown the kinds of stimuli that composed the deck, and the experimenter told the child the relevant dimension for that sort. The deck was shuffled and placed face-up in a card holder in front of the child. The child then picked up each card individually and placed it in a box to the left or to the right of the deck. The subject was permitted to look through the deck before beginning the sort, and to choose on which side (i.e., left or right) to place the exemplars that defined the classification. Each subject was allowed to pick up each card in any manner, but no subject was permitted to hold the deck of cards while sorting. Cards were not visible after a child had sorted them.

The sorting times were recorded by a stopwatch to the nearest 0.2 s. Errors were recorded, but no feedback on errors occurred. Each child was given frequent encouragement to sort quickly throughout the sessions. Each child was tested twice and received a small reward at the end of the second session. The experimenter was not aware of the expected performances of the children.

Design. The experiment followed a mixed factorial design. The three between factors included type of dimensional set (integrated vs separated), discriminability (high vs low), and age (kindergarten, second-grade, and fifth-grade children). Twelve subjects at each age were assigned to each stimulus set and level of discriminability. The within factors were target dimension and type of stimulus deck. Each child was required to sort each type of deck using each dimension of a set as the target. These six sorts constituted a single run. For example, a run for color vs orientation included (1) orientation control, (2) orientation correlated, (3) orientation orthogonal, (4) color control, (5) color correlated, and (6) color orthogonal.

Each subject was given six runs in two daily 45-min sessions. Runs 1–3 occurred on Day 1, and 4–6 on Day 2. The order of the sorts within a run was determined by a Latin square, and two subjects were assigned

to each order. One Latin square was used for Runs 1, 3, and 5, and its reverse was used for Runs 2, 4, and 6.

Results

Preliminary analyses were made to look for order effects and sex differences. These analyses revealed no significant effects, and these factors were not considered further. The errors averaged only 8.2%, and no formal analysis of the error data was attempted. It is important to note, however, that higher error rates were consistently associated with tasks that required a longer sorting time. Thus, no substantial speed/accuracy trade-off is involved in the experiment.

The mean sorting times, the dependent measure of primary interest, are presented in Tables 1 and 2 for the integrated and separated dimensions, respectively. Four separate ANOVAs were run to test the experimental hypotheses. Each type of dimensional combination was analyzed separately, and within each type of dimensional combination, separate analyses compared one dimension vs correlated-dimensions tasks and one-dimensional vs orthogonal-dimensions tasks. In each case, the ANOVA for a mixed factorial design was run. The between factors were Age (kindergarten, 2d grade, and 5th grade) and Discriminability (high vs low). The within factors were Task (one-dimension vs correlated dimensions or one-dimension vs orthogonal dimensions), Dimensions (color vs orientation or color vs number of lines). Following significant *F* tests, comparisons of specific means were made by comparing critical mean differences by the Bonferroni test (Myers, 1979) or correlated *t* tests. Unless otherwise stated the level of risk was .05.

TABLE 1
MEAN SORTING TIMES FOR INTEGRATED STIMULI

Discriminability	Grade	Dimension sorted	Type of deck		
			Correlated dimensions	One dimension	Orthogonal dimensions
Low	K	Angle	45.2	47.8	52.2
		Color	44.7	46.3	50.8
	2d	Angle	39.9	41.4	43.1
		Color	40.1	40.0	40.7
	5th	Angle	27.9	28.1	29.5
		Color	27.9	27.9	27.8
High	K	Angle	38.5	42.2	47.3
		Color	37.8	40.1	45.4
	2d	Angle	36.7	36.8	38.3
		Color	35.2	34.9	36.0
	5th	Angle	26.1	26.4	28.2
		Color	26.3	26.2	26.0

TABLE 2
MEAN SORTING TIMES FOR SEPARATED STIMULI

Discriminability	Grade	Type of deck			
		Dimension sorted	Correlated dimensions	One dimension	Orthogonal dimensions
Low	K	Lines	43.3	43.1	48.3
		Color	43.4	43.9	45.9
	2d	Lines	38.0	38.0	40.2
		Color	38.4	39.2	41.1
	5th	Lines	26.4	26.6	26.4
		Color	26.4	26.7	26.8
High	K	Lines	39.7	39.4	44.2
		Color	39.9	40.5	46.2
	2d	Lines	34.5	34.3	36.5
		Color	34.8	35.4	37.9
	5th	Lines	25.1	25.0	25.2
		Color	25.0	25.4	25.3

Control vs correlated dimensions. An inspection of the means in Table 1 indicates several obvious performance differences with spatially integrated dimensions that were confirmed by the ANOVA. There was an Age effect, $F(2, 66) = 168.96, p < .001$, and a Discriminability effect, $F(1, 66) = 34.13, p < .001$. Clearly, the mean sorting time varied inversely with age, and tasks with stimuli of high discriminability were easier than were tasks with stimuli of low discriminability. There was also an Age $\times$ Discriminability interaction, $F(2, 66) = 3.48, p < .05$, that indicated that the mean differences in sorting times between the high and low discriminability conditions decrease with increasing age.

There was also an effect due to Dimension, $F(1, 66) = 40.78, p < .001$, which reflects the finding that tasks with angle as the target dimension were more difficult than tasks with color as the target. There were also interactions due to Dimension $\times$ Task, $F(1, 66) = 28.34, p < .001$, Dimensions $\times$ Age, $F(2, 66) = 9.10, p < .001$, and an interaction of Dimension $\times$ Task $\times$ Age, $F(2, 66) = 3.19, p < .05$. The nature of these effects is shown by an examination of the second-order interaction (critical difference = 0.41). Kindergarten children sorted the correlated decks faster than the one-dimension decks regardless of the target dimension; fifth-graders did not sort correlated decks faster than control decks with either dimension as the target. Second-graders, on the other hand, sorted the correlated deck faster than the control deck when angle was the relevant dimension but not when color was relevant.

The theoretically interesting effects are shown by a main effect of Task, $F(1, 66) = 88.94, p < .001$, and interactions of Task $\times$ Age,

$F(2, 66) = 56.76, p < .001$, and Task $\times$ Age $\times$ Discriminability, $F(2, 66) = 4.70, p < .025$. Tests for simple effects (critical difference = 0.73) showed that kindergartners sorted correlated decks faster than control decks in both discriminability conditions; second-graders showed a redundancy gain in the low-discriminability condition but none in the high-discriminability condition. Although fifth-graders sorted significantly more slowly in the low-discriminability condition than in the high-discriminability condition, they showed no redundancy gain in either condition.

An additional analysis was conducted to determine whether the facilitation shown with correlated decks was due to a strategy that Garner (1974) dubbed "selective serial processing." A redundancy gain can occur in situations where one dimension is harder than another, because subjects sort the correlated decks using the easier of the two dimensions. The detection of this strategy may not be possible in an overall analysis if different dimensions are more difficult for different subjects. In order to check for selective serial processing, the performances of individual subjects in each age group were compared. The mean sorting time for the correlated dimensions decks was compared to the sorting time for the faster one-dimension deck using correlated t tests. For kindergartners, the mean sorting time of the correlated-dimensions deck was significantly faster than the sorting of the faster one-dimension deck, $t(23) = 7.44, p < .001$, but the mean time for the correlated-dimensions deck was not faster than the one-dimension deck for either the second- or fifth-grade children. Thus, the redundancy gain for the kindergarten children is genuine, but that shown by the older groups of children is due to selective serial processing. The latter conclusion was further confirmed by comparing the mean sorting time of the correlated-dimensions decks to the sorting time of the slower control. Both of the older age groups sorted the correlated-dimensions decks faster than the slower control (second-grade: $t(23) = 6.08, p < .01$; fifth-grade: $t(23) = 3.58, p < .01$).

It should be noted that the interaction between age and task cannot be attributed to a floor effect. Fifth-graders, for example, did sort more slowly in the low-discriminability condition than they did in the high-discriminability condition. In principle, then, a redundancy gain could have been shown by these children, in the low-discriminability condition, at least, if the stimuli were perceived holistically.

Consider now the performance with the separated dimensional combinations. An inspection of Table 2 indicates a pronounced effect of age and discriminability. This impression was confirmed by the results of the ANOVA which showed a main effect of Age, $F(2, 66) = 194.44, p < .001$, and Discriminability, $F(1, 66) = 18.02, p < .001$. Subsequent comparisons of the means indicated that the performance level of each age group differed from each of the other age groups, and stimuli of high

discriminability were sorted faster than stimuli of low discriminability. The interaction between age and discriminability was not reliable.

The important differences were shown by a main effect of Task, $F(1, 66) = 11.89$, $p < .001$; Dimension, $F(1, 66) = 15.96$, $p < .001$; and a Task $\times$ Dimension interaction, $F(1, 66) = 12.63$, $p < .001$. As shown in Table 2, tasks with color as the relevant dimension took longer than tasks with the number of lines as the relevant dimension. Although tasks with correlated dimensions were faster than tasks with one dimension, this effect occurred only when color was the target dimension (critical difference = 0.32).

For each age group, further comparisons were made to check for selective serial processing. At each developmental level for both conditions of discriminability, the mean sorting time for the correlated-dimensions task was shorter than the slower of the one-dimension tasks (all $ts > 3.4$, $p < .01$), but not the faster of the one-dimension tasks, $p > .05$. It is of particular interest to note that although the redundancy gain shown by kindergarten children with integrated stimuli can be attributed to the holistic structure of integral stimuli, the redundancy gain shown by kindergartners with correlated separated dimensions can be only attributed to selective serial processing.

Control vs Orthogonal Dimensions. An inspection of the means in Table 1 reveals several strong effects associated with the spatially integrated dimensions. An ANOVA shows main effects due to Age, $F(2, 66) = 205.27$, $p < .001$, Discriminability, $F(1, 66) = 27.60$, $p < .001$, Task, $F(1, 66) = 250.97$, $p < .001$, and Dimension, $F(1, 66) = 88.82$, $p < .001$. There were also significant first-order interactions due to Task $\times$ Dimension, $F(1, 66) = 16.10$, $p < .001$, and Task $\times$ Age, $F(2, 66) = 93.94$, $p < .001$.

Although these effects seemed somewhat complicated at first glance, all effects, but one, were readily interpretable by considering the significant second-order Task $\times$ Dimension $\times$ Age interaction, $F(2, 66) = 8.25$, $p < .001$. The exception is the effect of Discriminability. The stimuli of low discriminability took longer to sort than stimuli of high discriminability, but discriminability did not interact with any other factor.

The Task $\times$ Dimension $\times$ Age interaction is simple to interpret. Recall that the critical comparison is between control and orthogonal decks in which the same dimension serves as the basis for the sort. For kindergarten children, the orthogonal-dimensions task took longer than the control, and the effect was equally potent for both dimensions. For second-grade children, the interference effect due to orthogonal dimensions was substantially less than for the kindergarten children, but again the interference was produced with both dimensions. For fifth-grade children, an asymmetric interference effect was observed. When angle was the target dimension, orthogonal variations of color produced strong interference. When color

was the target dimension, however, orthogonal variations of angle produced no reliable interference (critical difference = 0.84).

Consider now the performances on the spatially separated dimensions. An ANOVA of mean sorting times showed main effects due to Age, $F(2, 66) = 245.24$, $p < .001$, and Discriminability, $F(1, 66) = 16.49$, $p < .001$. There was a clear inverse relation between Age and sorting time, and stimuli of low discriminability took longer to sort than did stimuli of high discriminability.

There is also a main effect of Dimension, $F(1, 66) = 22.46$, $p < .001$, and a weak interaction between Dimension and Age, $F(2, 66) = 3.45$, $p < .05$. Tests for simple effects (critical difference = 0.82) showed that tasks with color as the target dimension were more difficult than tasks with number of lines as the relevant dimension for both the kindergarten and the second-grade children, but there were no differences between the two dimensions for fifth-grade children.

The critical effects are shown by a main effect of Task, $F(1, 66) = 250.97$, $p < .001$, and a significant Task $\times$ Age interaction, $F(2, 66) = 93.94$, $p < .001$. Tests for simple effects showed an inverse relation between age and the amount of interference. For both kindergarten and second-grade children, there were reliable differences between the sorting times of control and orthogonal tasks, but for the fifth-grade children, the differences were not reliable (critical difference = 0.39).

Note once again that the Age $\times$ Task interaction cannot be attributed to a floor effect. The fifth-grade children showed no interference with orthogonal dimension tasks in either the low- or high-discriminability conditions, which were sorted at significantly different speeds. Thus, the equivalent performances of these subjects in the control and orthogonal dimensions tasks with stimuli of low discriminability, at least, is attributable to the subjects' ability to attend selectively rather than an inability to sort the cards any faster in the control task.

DISCUSSION

The results of this experiment strongly support the hypothesis that there are developmental differences in perceived structure as well as differences in the ability to allocate attentional resources. Consider first the performances with correlated dimensions. With the spatially separated dimensions, color vs number of lines, the speed of sorting on correlated dimensions relative to the sorting on single dimensions showed no redundancy gains at any developmental level other than gains that could be attributed to selective serial processing. The allowable inference in clear: The dimensions are perceptually independent at the ages we tested.

A different picture emerges for the integrated dimensions. A comparison of sorting times for correlated values of color vs angle with the times for single dimensions revealed a genuine redundancy gain for kinder-

gartners, but only selective serial processing for second- and fifth-graders. Thus, combinations of these dimensions are perceived as integral stimuli by younger children and as conjunctions of perceptually independent features by older children.

The performances of subjects with correlated dimensions clarifies the interpretation of performances with orthogonal dimensions when selective attention is required. If the dimensions are perceptually independent, interference effects may be attributed directly to an inability to allocate attention resources. If, on the other hand, perceptual dependence exists between dimensions, interference effects depend jointly upon perceived structure and attention.

The operation of attention is clearly characterized with spatially separated stimuli. Each stimulus is a perceptual unit toward which attention can be directed. Attention can move to the location of a specific target (e.g., Tsal, 1983), and when attention is deployed efficiently, distraction from spatially separated irrelevant stimuli should be minimal. Attentional deployment would be expected to improve with development, and subjects would be less distracted by irrelevant information with increasing age (see, e.g., Barrett & Shepp, 1985). Younger children, with less efficient deployment than older subjects, would be likely to make an obligatory check on the irrelevant stimulus and interference would occur. In contrast, older children, with efficient deployment of attention, would be unlikely to be so distracted. This type of interpretation is similar to one offered by Well et al. (1980), and, like their view, argues that the time that attention is captured by irrelevant stimuli is independent of the time taken to process the relevant dimension. Such an argument calls for an additive model of interference, where the amount of interference is independent of the difficulty of the baseline or control task. The additive model is supported by the observation that the amount of interference due to orthogonal dimensions is the same at each developmental level, regardless of task difficulty.

Attention to spatially integrated dimensions is difficult to characterize because of the complex relations between attention and perceived structure, and because these relations change with the ages of the observer. Consider the performances of kindergartners. These children perceived the stimuli as integral and interference occurred in the selective attention task. But the interference could arise in either of two ways. According to one, integral stimuli are unitary percepts: there are no parts to which selective attention can be directed. Attention is directed to the holistic stimuli, and the classification is accomplished by overall similarity relations. The control task is performed by mapping each of the two types of stimuli to one of the responses. The orthogonal dimension task shows interference because there are four types of stimuli; two must be mapped to one response and two must be mapped to the other.

Interference with integral stimuli can also occur when the subject analyzes an integral whole into its constituent features. According to Lockhead (1972) such an analysis is possible, but occurs only when called for by the demands of the task. In the context of the speeded sorting task, an analysis of the stimuli would occur only in some conditions. The control task is easily accomplished by the classification of holistic stimuli; the orthogonal-dimensions task, on the other hand, can be managed more accurately if the stimuli are classified by constituent features. In the latter case, the whole would be analyzed into its features, and the additional operation would take more time relative to the control task.

The results obtained with kindergartners are consistent with either the view of stimulus-response mapping or the analysis-by-feature alternative. Comparisons of the mean sorting times of the control tasks with the orthogonal dimensions tasks reveals that the amount of interference in the high-discriminability condition is equivalent to that shown in the low-discriminability condition. This finding is consistent with an additive model of interference (Well *et al.*, 1980) and can be predicted by either view. Whether the child engages in response mapping or stimulus analysis depends, in all likelihood, on both the child's perception of the task and the analyzability of the stimuli.

The performances of the second- and fifth-graders follow a pattern that departs from the usual integrality to separability trend in perceptual development (e.g., Shepp & Swartz, 1976). Although the comparisons between correlated-dimensions and control tasks indicated that color vs angle are perceived as subjectively independent dimensions, there is also clear evidence of performance asymmetry—a finding that is also present in comparisons of the orthogonal-dimensions and control tasks.

Twenty-two of 24 second-graders found color to be an easier dimension than orientation. In the correlated-dimensions task there was a redundancy gain when orientation was the target dimension but not when color was the target dimension. A similar asymmetry appeared with orthogonal dimensions: irrelevant variations of color produced more interference than did irrelevant variations of orientation.

Fifth-graders also showed a performance asymmetry, but it was not dependent on one dimension being easier than the other. Overall, color was the easier dimension for only 12 subjects, and selective serial processing with correlated dimensions was associated with the easier discrimination rather than with a particular dimension consistently. With orthogonal dimensions, however, there was a clear pattern of asymmetric interference: irrelevant variations in color significantly interfered with the classification of orientation, whereas irrelevant variations in orientation produced no significant interference in the classification of color. Possible interpretations of this asymmetry are considered later.

EXPERIMENT 2

The results of Experiment 1 are consistent with the claim that dimensional combinations that are perceived by older children and adults as conjunctions of separable features are perceived by young children as integral wholes (Shepp & Swartz, 1976; Shepp, 1978; Smith & Kemler, 1977, 1978). These results also indicate that the development of perceived structure can take more than a single form. The appearance of asymmetric interference with spatially integral dimensions in older children suggests that the features of some dimensions do not achieve complete perceptual independence and may be initially perceived as holistic. This hypothesis is consistent with the view of Lockhead (1972), and the parallel developmental view proposed by Shepp (1978), that integral stimuli are initially perceived as holistic and are only subsequently decomposed into specific dimensional features. Support for this view has been provided by Shepp (1986) who reports that size and shape are processed holistically in a divided attention task even though they are asymmetrically analyzed by features in an unspeeded restricted classification task.

According to this view of a holistic to featural progression, the integration of multiple sources of information undoubtedly depends on the types of dimensional interactions that occur in the stimulus (Garner, 1974, 1976; Lockhead, 1972, 1979) as well as those that are yielded by the mechanisms of preattentive grouping (Kahnemann, 1973). For either alternative, a theoretical analysis must begin with the identification of those aspects of the stimulus that promote the integration or differentiation of information.

Two studies have shown that spatial and form similarity contribute to the integration of information. In the first, Pomerantz and Schwaizberg (1975), using measures of selective and divided attention, showed that parentheses are perceived holistically when the left and right parentheses of a pair are spatially continuous, but are perceived according to features as the spatial distance between the parentheses is increased. A similar result for aspects of shape has been reported by Garner (1978). Garner's results show that two continuous parallel lines are perceived holistically, but when one of the lines of the pair is rotated 90°, the featural rather than the holistic properties of the stimuli control performance.

In the studies by Pomerantz and Schwaizberg (1975) and Garner (1978), the color of the elements was constant, which suggests that perceptual independence between elements can exist even though elements share an identical color. It is plausible, however, that under some conditions, common colors can facilitate the integration of information and that distinctive colors can facilitate the differentiation of features. As shown in Fig. 1, the perceptual dependence between color and angle that was shown by the results of Experiment 1 could have been affected by the uniform color of the entire configuration. If so, perceptual independence

could be promoted by arranging one color on the circle and another on the pointer that displays the angle of inclination. Under these conditions, the perceptual dependencies, symmetric as well as asymmetric, that were observed in Experiment 1 should be eliminated.

Method

Subjects. The subjects were 12 kindergartners and 12 fifth-graders who were attending school in East Providence, Rhode Island. The mean age of the kindergartners was 5 years, 2 months (range: 4 years, 10 months to 5 years, 9 months). The mean age of the fifth-graders was 11 years, 7 months (range: 11 years to 12 years, 1 month).

Stimuli. The color and angle dimensions that are illustrated in Fig. 1 were modified by making the center circle one color and the pointer a second color. In changing the colors of these spatially adjacent areas, both the contour and the color of the stimuli were changed. The color discrimination, red vs orange-red, was based upon the colors of two filled circles. The pointers were colored black and were inclined at 30° and 330° from the vertical. In other respects, the stimuli were the same as in Experiment 1.

Task and procedure. The task and procedure were the same as in Experiment 1.

Design. The experiment was run as a mixed-factorial design. There was one between variable, Age (kindergarten vs fifth-grade) and two within variables, Dimension (color vs angle) and Task (control vs correlated vs orthogonal). Other characteristics of the design were the same as in Experiment 1.

Results and Discussion

Preliminary analyses revealed no order effects or sex differences in sorting times or errors, and these factors were considered no further. The errors averaged only 7.7%, and no formal analysis of the error data was performed. As in Experiment 1, higher error rates were consistently associated with tasks that required longer sorting times.

The mean sorting times are shown in Table 3. The means for correlated-

TABLE 3
MEAN SORTING TIMES FOR SPATIALLY DISTINCT COLOR AND ANGLE STIMULI

Grade	Dimension sorted	Correlated dimensions	One dimension	Orthogonal dimensions
K	Angle	40.0	40.2	44.9
	Color	38.8	38.7	42.9
5th	Angle	26.7	27.2	26.9
	Color	26.9	26.8	26.8

dimensions and orthogonal-dimensions tasks were compared to the control task by separate ANOVAs for a mixed-factorial design. In each analysis, the between factor was Age (kindergarten vs fifth-grade), whereas the within factors were Task (control vs correlated dimensions or control vs orthogonal dimensions) and Dimension (color vs angle).

A comparison of the sorting times between the control and correlated-dimensions tasks revealed a main effect of Age, $F(1, 22) = 88.96$, $p < .001$, Dimensions, $F(1, 22) = 15.89$, $p < .001$, and an interaction between Dimensions and Age, $F(1, 22) = 16.31$, $p < .001$. Overall, the kindergarten children sorted more slowly than did the fifth-graders, and tasks that called for a classification by angle were performed more slowly than were tasks that were classified by color. The latter conclusion holds only for kindergarten children, and is the source of the interaction between age and dimensions. The mean sorting times on the two dimensions do not differ for fifth-graders (critical difference = 0.62).

There was also a main effect due to Task, $F(1, 22) = 4.54$, $p < .05$, but an inspection of Table 3 reveals no appreciable facilitation due to correlated dimensions. A comparison of the mean sorting times of the correlated-dimensions tasks with the mean of the faster control task, using correlated t tests, revealed that the difference was not significant ($p > .05$) for either the kindergarten or fifth-grade children. A similar comparison of the sorting times of the correlated-dimensions task with the slower control task, however, revealed that the former was faster than the latter for both kindergarten, $t(11) = 3.98$, $p < .01$, and fifth-grade children, $t(11) = 4.86$, $p < .01$. Thus, although there is no redundancy gain that can be attributed to the presence of integral stimuli, there is clear evidence of selective serial processing.

Consider now comparisons between the control and the orthogonal-dimensions tasks. Significant effects were observed for Age, $F(1, 22) = 123.94$, $p < .001$, Dimension, $F(1, 22) = 55.22$, $p < .001$, and the Age $\times$ Dimension interaction, $F(1, 22) = 40.20$, $p < .001$. As in the preceding analysis, angle was more difficult than color, but the effect was confined to the kindergarten children who generally sorted more slowly (critical difference = 0.47). Significant effects due to Task, $F(1, 22) = 415.32$, $p < .001$, and the interaction between Task and Age, $F(1, 22) = 524.01$, $p < .001$ were also observed. As shown in Table 3, the performance of kindergarten children shows substantial interference with orthogonal dimensions, but there was no interference observed in the performance of fifth-grade children (critical difference = 0.40). Taken together, the results of Experiment 2 show that the dimensions of angle and color, which appear to be aspects of a single holistic object in Experiment 1, can be separated perceptually by making the color of the pointer different from that of the central region of the figure. This operation has the effect of making the stimuli separable for children of both age levels, but it does

not eliminate the difficulty young children experience when attempting to attend selectively to a specific target dimension.

GENERAL DISCUSSION

The results of the present experiments bear on two current theoretical issues in perceptual development. The issues include (1) the development of perceived structure and the relation of perceived structure to attention, and (2) alternative views concerning the development of perceived structure.

The Development of Perceived Structure and Attention

Several recent investigations (Shepp, 1978; Shepp & Swartz, 1976; Smith & Kemler, 1977, 1978) have shown that dimensional combinations that are perceived by older children and adults as separable are perceived by younger children as integral. Such results have led these investigators to assume that holistic perception is the preferred or dominant mode of organization early in development, and that with increasing age and experience, older children and adults become increasingly adept in accessing the dimensional relations that characterize a set of multidimensional stimuli.

The present results support and extend the argument in several interesting ways. The performances of the kindergarten children indicate that color and orientation are perceived as integral, and there is no evidence that the stimuli are analyzed according to specific features. Fifth-grade children, however, do appear to analyze the stimuli according to specific features; no redundancy gain appears with correlated values, a result that also occurs with spatially separate stimuli. Moreover, the asymmetric interference shown by the fifth-graders on tasks with orthogonal dimensions is a result that can occur only if the stimuli are analyzed.

A developmental trend in the direction of an asymmetric separable organization is a departure from the pattern of results previously observed with similar stimuli (e.g., Shepp & Swartz, 1976; Smith & Kemler, 1978) and requires further consideration. Two general views about perceptual asymmetries have been offered: one implicates levels of processing, whereas the other emphasizes different rates of processing.

Wood (1974, 1975) has suggested that perceptual asymmetries may result from different levels of processing. In Wood's experiments subjects performed speeded classifications in which the target stimuli either varied in pitch, a nonlinguistic dimension, or varied in consonant sound, a linguistic dimension. He observed that orthogonal variations in the consonants did not interfere with the classification of pitch, but that orthogonal variations in pitch interfere with classifications based on the consonants. Wood argues that pitch is processed at a lower or earlier level than the linguistic consonant. Thus, pitch would be processed and a response generated before irrelevant variations in the consonants could interfere

with processing. The processing of consonants, however, would require the prior processing of pitch that would then interfere with the classification.

Color and orientation are not readily classified as nonlinguistic and linguistic dimensions, respectively, but detector systems may be organized in a variety of sequentially arranged levels such that one dimension is registered prior to another. Moreover, there is evidence to suggest that color is more available than orientation early in processing (von Wright, 1972), and the early availability of color would yield an asymmetry in processing.

Perceptual asymmetries can also arise from differences in the rate of processing. Posner (1969, 1978; Posner & Mitchell, 1967) has argued that the features of multidimensional stimuli are encoded in parallel but that the recognition of the features may be accomplished at different rates. If the classification is based on the faster of the two features, the item can be recognized and the response initiated before the processing of the second, interfering feature is completed. Under these conditions, no interference will occur. If, on the other hand, classification is based on the slower feature, the item will be recognized subsequent to the processing and recognition of the faster feature. When the two features compete for the selection of a response, interference will occur.

These views of perceptual organization are compatible with each other, and are consistent with recent views that speak to the development of perceptual organization (e.g., Shepp, 1978; Smith & Kemler, 1978). With increasing age some internal representations may become independent of others and arranged as either hierarchies or levels. The child may also become either more adept at initiating a response as soon as the necessary information is available or more efficient in reading out available sources of information, some aspects of which are processed more rapidly than others.

The results of Experiments 1 and 2 also clarify the nature of the relation between the development of perceived structure and the development of attention. When features are integral, the stimuli are perceived holistically, and it is presumably the nature of the perceived structure rather than any resource deficiency that leads to a failure of selective attention. Clearly, however, some integral stimuli are easier to analyze into constituent features than others. Hue and chroma of a Munsell chip, for example, are very difficult to analyze, and the accuracy in identifying specific features is very poor (Shepp, 1983); in speeded sorting, orthogonal dimensions with hue vs chroma produce mutual interference in children and adults alike. These kinds of dimensions do not permit selective attention: interference effects are attributable to perceived structure.

Other stimuli, however, such as color and orientation are analyzable. With increasing age there is a loss of a redundancy gain with correlated values, and an increasing performance asymmetry in tasks that require

selection attention. The later finding, in particular, indicates that the developing child becomes more effective in analyzing the features of a stimulus as well as acquiring increasing command of attentional resources. Shepp (1986) has shown a similar effect with size and shape that extends into adulthood.

The performances of children in tasks with spatially separate dimensions provides still another look at perceptual development. Color of circle provides a source of information that is independent of the number of lines in an external ring (Experiment 1) or the inclination of a pointer that bears a different color (Experiment 2). These dimensional combinations appear to be separable for subjects, regardless of age; no redundancy gain was observed with correlated values at any age. But there was clearly interference in the performance of kindergartners with orthogonal dimensions; the interference declined to a minimum as age increased, and it occurred in easy as well as hard tasks. These results support the conclusion that children acquire increasing command of attentional resources with increasing age.

Alternative Views of Perceptual and Cognitive Development

The view that perceptual development proceeds from object perception that is undifferentiated or holistic to object perception that is differentiated or articulated by dimensional features is an old one (Bruner, Olver, & Greenfield, 1964; Gibson, 1969; Werner, 1948). The characterization of this perceptual trend in terms of the distinction between integral and separable dimensions is clearly consistent with these early formulations, but it can be considered as a more specific formulation of the issue and one that provides a variety of converging operations to test its validity. Moreover, this alternative characterization of perceptual development has led to some very novel and promising lines of developmental research (e.g., Kemler, 1983a; Smith, 1979).

Recently, the holistic to featural characterization of perceptual development has been challenged by Odom and his colleagues (e.g., Aschkenasy & Odom, 1982) who offer an alternative view. Aschkenasy and Odom argue that during the course of development children tend to discover more and more dimensions in multidimensional objects, and are more likely to detect smaller and smaller differences between stimuli on any given dimension. According to this argument, the performances of children in classification and learning tasks depend upon the dimensions that exist in the hierarchies at a particular age, the relative salience of these dimensions, and the ways in which the dimensional hierarchies of a particular developmental level are matched by the experimenter-defined dimensions of a particular task. The view proposed by Aschkenasy and Odom is not incompatible with the separability hypothesis (Shepp, 1978) and, as Kemler (1983b) has pointed out, the data from their experiments

are literally predicted by the separability hypothesis. Kemler also points out quite clearly and eloquently some of the implications of the separability hypothesis and the contributions of investigators that have been developed within this framework.

We need not repeat Kemler's arguments here. Some aspects of our results, however, bear directly upon the comparison between the separability hypothesis and the view proposed by Aschkenasy and Odom (1982). Their arguments about the importance of dimensional organizations are addressed primarily to the results of learning and classification tasks. The separability hypothesis also speaks to the performances in such tasks, but the hypothesis is supported by evidence for other converging tasks as well. These converging operations include the speeded sorting tasks of the present study, and it is with more than a passing interest that we ask how well the Aschkenasy and Odom view deals with such results. Although their view does allow for developmental differences in sensitivity to particular dimensions, it does not posit that performances differ because of differences in perceived dimensional structure. Thus, developmental differences in performance that occur on either the orthogonal- or correlated-dimensions tasks relative to the control task should depend exclusively on dimensional salience effects and how such effects change with age.

Aschkenasy and Odom's model is but one of a class of models that take dimensions as the organizational unit of the stimulus, and certain types of redundancy gains and interference effects can be predicted by these models (cf. Lockhead, 1972). But such models presently fail in two important ways. First, the models predict similar patterns of performance in tasks with integral and separable dimensions; our results argue strongly against such a prediction. Second, as Lockhead (1979) has forcefully argued, predictions for one kind of task (e.g., correlated values) by dimensional models typically fail even though the specific stimuli are known to obey a dimensional organization. These difficulties lead us to believe that it is shortsighted to argue that holistic and dimensional organizations are opposing in nature, particularly when such a broad range of results can be explained by a view that incorporates both forms of organization.

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